Inclusive method

In this method, both the lower and upper class limits are included in the classes. Inclusive type of classification may be used for a grouped frequency distribution for discrete variable like members in a family, number of workers etc., It cannot be used in the case of continuous variable like height, weight etc., where integral as well as fractional values are permissible. Since both upper limit and lower limit of classes are included for frequency calculation, this method is called inclusive method.

Exclusive method

In this method, the values which are equal to upper limit of a class are not included in that class and instead they would be included in the next class. The upper limit is not at all taken into consideration or in other words it is always excluded from the consideration. Hence this method is called exclusive method .

Example 3.13

The marks scored by 50 students in an examination are given as follows:

23, 25, 36, 39, 37, 41, 42, 22, 26, 35, 34, 30, 29, 27, 47, 40, 31, 32, 43, 45, 34, 46, 23, 24, 27, 36, 41, 43, 39, 38, 28, 32, 42, 33, 46, 23, 34, 41, 40, 30, 45, 42, 39, 37, 38, 42, 44, 46, 29, 37.

It can be observed from this data set that the marks of 50 students vary from 22 to 47. If it is decided to divide this group into 6 smaller groups, we can have the boundary lines fixed as 25, 30, 35, 40, 45 and 50 marks. Then, we form the six groups with the boundaries as 21 - 25, 26 - 30, 31 - 35, 36 - 40, 41 – 45 and 46 - 50.

The continuous frequency distribution formed by inclusive and exclusive methods are displayed in Table 3.13(i) and Table 3.13(ii), respectively.

Table 3.13(i)

Marks secured by students (Inclusive Method)

Marks	x, Integer value	Tally Marks	No. of Students
21-25	$21 \leq x \leq 25$	𝍸 /	6
26-30	$26 \leq x \leq 30$	𝍸 ///	8
31-35	$31 \leq x \leq 35$	𝍸 ///	8
36-40	$36 \leq x \leq 40$	𝍸 𝍸 //	12
41-45	$41 \leq x \leq 45$	𝍸 𝍸 //	12
46-50	$46 \leq x \leq 50$	////	4
		Total	**50**

Ch3_11th Stats.indd 57 2/3/2019 10:58:16 AM